

CHAPTER - TWO

STRIP METHOD FOR SLAB DESIGN

Introduction

The upper bound theorem of the theory of plasticity was present in yield line theory. The yield line method of slab analysis is an upper bound approach to determine the capacity of slabs.

Disadvantage:

- In upper bound analysis if an error occurs it will be on the unsafe side. The actual carrying capacity will be less than, or at best equal to the capacity predicted, which is certainly a cause for concern in design.
- When applying this method it necessary to assume the distribution of reinforcement is known over the whole slab. It can be used for design only in an iterative sense i.e. trial design until a satisfactory arrangement is found.

These circumstances motivated Hillerborg (1956) to develop what is known as the strip method for slab design. In contrast to yield line analysis, the strip method is a lower bound approach, based on the satisfaction of equilibrium requirements everywhere in the slab. By the strip method, a moment field is first determined that fulfills equilibrium requirements, after which the reinforcement of the slab at each point is designed for this moment field.

Lower Bound Theorem

If a distribution of moment can be found that satisfies both equilibrium and boundary conditions for a given external loading, and if the yield moment capacity of the slab is no where exceeded, then the given external loading will represent a lower bound of the true carrying capacity.

Advantages:

- The strip method gives results on the safe side, which is certainly preferable in practice.
- The strip method is a design method by which the needed reinforcement can be calculated.

Basic Principles

The governing equilibrium equation for a small slab element having sides dx and dy is:

$$\frac{\partial^2 m_x}{\partial x^2} + \frac{\partial^2 m_y}{\partial y^2} - \frac{\partial^2 m_{xy}}{\partial x \partial y} = -W$$

Where: W = The external load per unit area.

m_x and m_y = BM's per unit width in x and y directions respectively.

m_{xy} = the twisting moment.

Hence, according to the lower bound theorem, any combination of m_x , m_y and m_{xy} that satisfies the equilibrium equation at all points in the slab and that meets boundary conditions is a valid solution, provided that the reinforcement is placed to carry these moments.

The basis for the simple strip method is that the torsional moment is chosen equal to zero; no load is assumed to be resisted by the twisting strength of the slab. (The reinforcements are parallel to the axes in the rectilinear coordinate system)

$$\therefore m_{xy} = 0$$

The equilibrium equation then reduces to

$$\frac{\partial^2 m_x}{\partial x^2} + \frac{\partial^2 m_y}{\partial y^2} = -W$$

This equation can be split conveniently into two parts, representing twist less beam strip action,

$$\frac{\partial^2 m_x}{\partial x^2} = -kW$$

$$\frac{\partial^2 m_y}{\partial y^2} = -(1-k)W$$

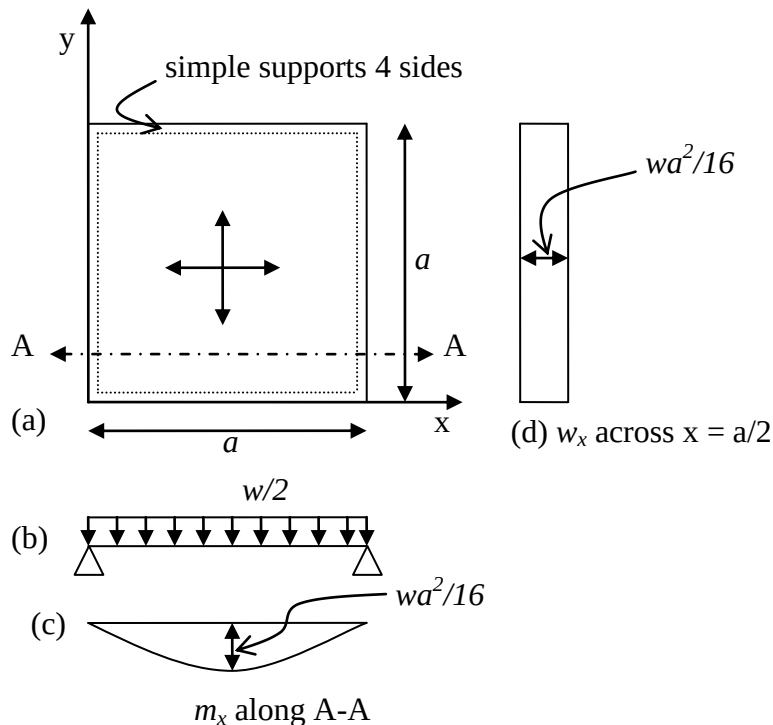
Where the proportion of the load taken by the strips is k in the x- direction and $(1-k)$ in the y- direction. In many regions in slabs, the value k will be either 0 or 1 i.e. load is dispersed by strips in x- or in y- direction. In other regions, it may be reasonable to assume that the load is divided equally in two directions (i.e. $k = 0.5$).

Choice of Load Distribution

Theoretically, the load W can be divided arbitrarily between x- and y- directions. Different divisions will of course, lead to different patterns of reinforcement, and all will not be equally appropriate. The desired goal is to an arrangement of steel that is safe and economical and that will avoid problems at the service load level associated with excessive cracking or deflections. Knowledge of the moment field

according to the elastic theory is thereby very helpful. In general, the designer may be guided by his knowledge of the general distribution of elastic moments.

To see an example of the strip method and to illustrate the choices open to the designer. Consider the square, simply supported slab shown below, with side length a and a uniformly distributed factored load w per unit area. The simplest load distribution is obtained by setting $k = 0.5$ over the entire slab, as shown.



The load on all strips in each direction is thus $w/2$ (with $k = 0.5$), as illustrated by the *load dispersion arrows*. This gives maximum design moments

$$m_x = m_y = wa^2/16$$

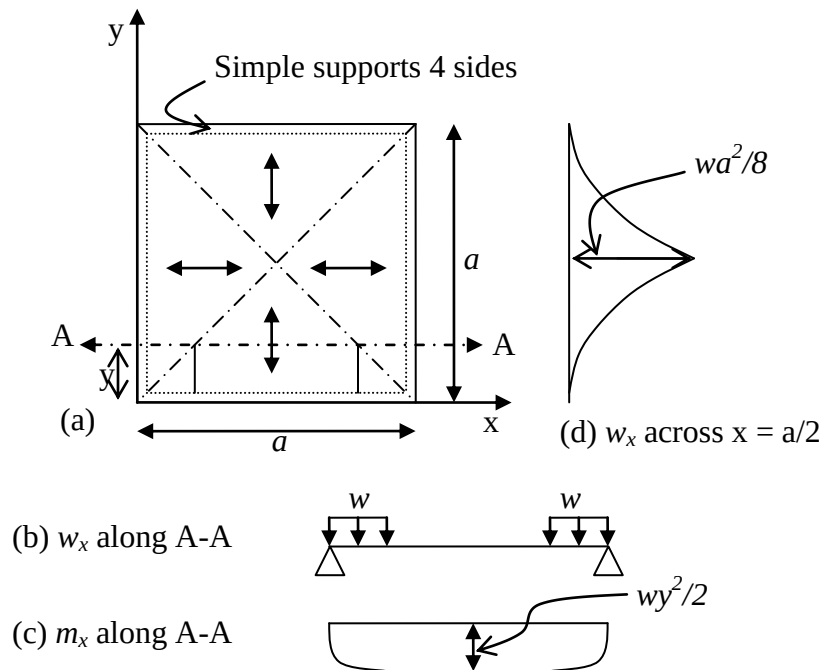
Implying a constant curvature for strips in the x - direction at $x = a/2$ corresponding to a constant moment $wa^2/16$ (see fig. d). Similar constant curvatures are also expected at various x 's corresponding to the constant BM's at $x = \text{constant}$. The same applies for y -direction strips.

It is recognized however that the curvatures, hence the moments, must be greater in the strips near the middle of the slab than near the edges. If the slab were reinforced according to this solution

extensive redistribution of moments would be required, certainly accompanied by much cracking in the highly stressed regions near the middle of the slab.

So what we need is a type of load distribution which can give a moment distribution such that we get great curvatures in say x- direction strips near slab middle and less near the edges.

Try the alternative, more reasonable distribution shown below. Here the regions of different load dispersion, separated by the dashed dotted discontinuity lines, follow the diagonals, all of the load on any region is carried in the direction giving the shortest distance to the nearest support ($k = 0$ or $k = 1$ in the different regions)



The lateral distribution of moments shown in figure (d) would theoretically require a continuously variable bar spacing, obviously impracticality. A practical solution would be to reinforce for the average moment over a certain width, approximating the actual lateral variation in figure (d) in a stepwise manner. Hillerborg notes that this is not strictly in accordance with equilibrium theory and that the design is no longer certainly on safe side, but other conservative assumptions, e.g., neglect of membrane

strength in the slab or strain hardening of the reinforcement, would surely compensate for the slight reduction in safety margin.

A third alternative is with discontinuity lines parallel to the edges. Here again the division load is made so that the load is carried to the nearest support, as before, but load near the diagonals has been divided, with one-half taken in each direction. Thus k is given values 0 or 1 along the middle edges and 0.5 in the corners and center of the slab, with load dispersion in the directions indicated by the arrows.

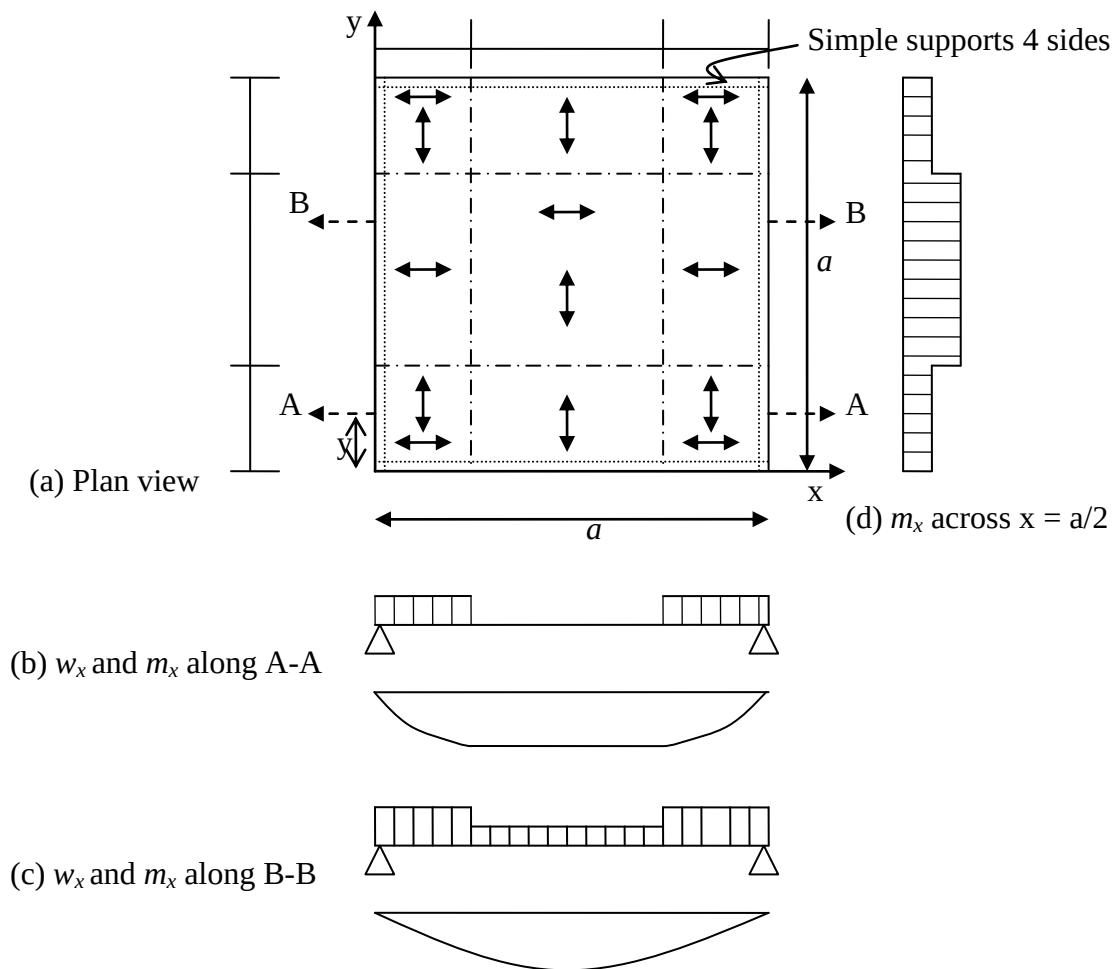


Fig. Square slab with load near diagonals shared equally in two directions

Two different strip loadings are now identified. For an x - direction strip along section A-A, the maximum moment is:

$$m_x = \frac{w}{2} * \frac{a}{4} * \frac{a}{8} = \frac{wa^2}{64}$$

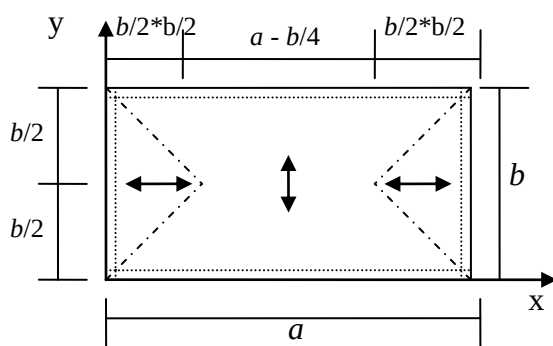
And for a strip along section B-B, the maximum moment is:

$$m_x = w * \frac{a}{4} * \frac{a}{8} + \frac{w}{2} * \frac{a}{4} * \frac{3a}{8} = \frac{5wa^2}{64}$$

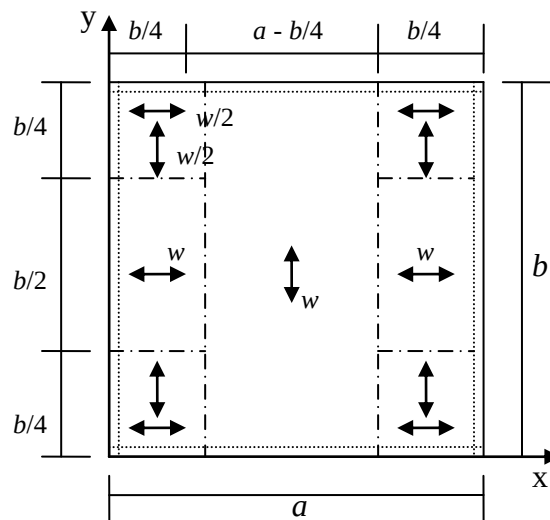
This design leads to a practical arrangement of reinforcement, one with constant spacing through the centre strip of width $a/2$ and a wider spacing through the outer strips, where the elastic curvatures and moments are known to be less. The averaging of moments necessitated in the second solution is avoided here, and third solution is fully consistent with the equilibrium theory.

The three examples also illustrate the simple way in which the moments in the slab can be found by the strip method, based on familiar beam analysis. It is important note, too, that the load on the supporting beams is easily found because it can be computed from the end reactions of the slab-beam strips in all cases.

Rectangular slabs



(b) Rectangular slab with discontinuity lines originating at corners



(c) Rectangular slab with discontinuity lines parallel to the sides

The second, preferred arrangement, shown in Fig. (b) gives design moments as follows:

In the x- direction

$$\text{Side strips: } m_x = w/2 * b/4 * b/8 = wb^2/64$$

$$\text{Middle strips: } m_x = w * b/4 * b/8 = wb^2/32$$

In the y- direction

$$\text{Side strips: } m_y = wb^2/64$$

$$\text{Middle strips: } m_y = wb^2/8$$

This distribution, requiring no averaging of moment across band widths, is always on the safe side and is both simple and economical.

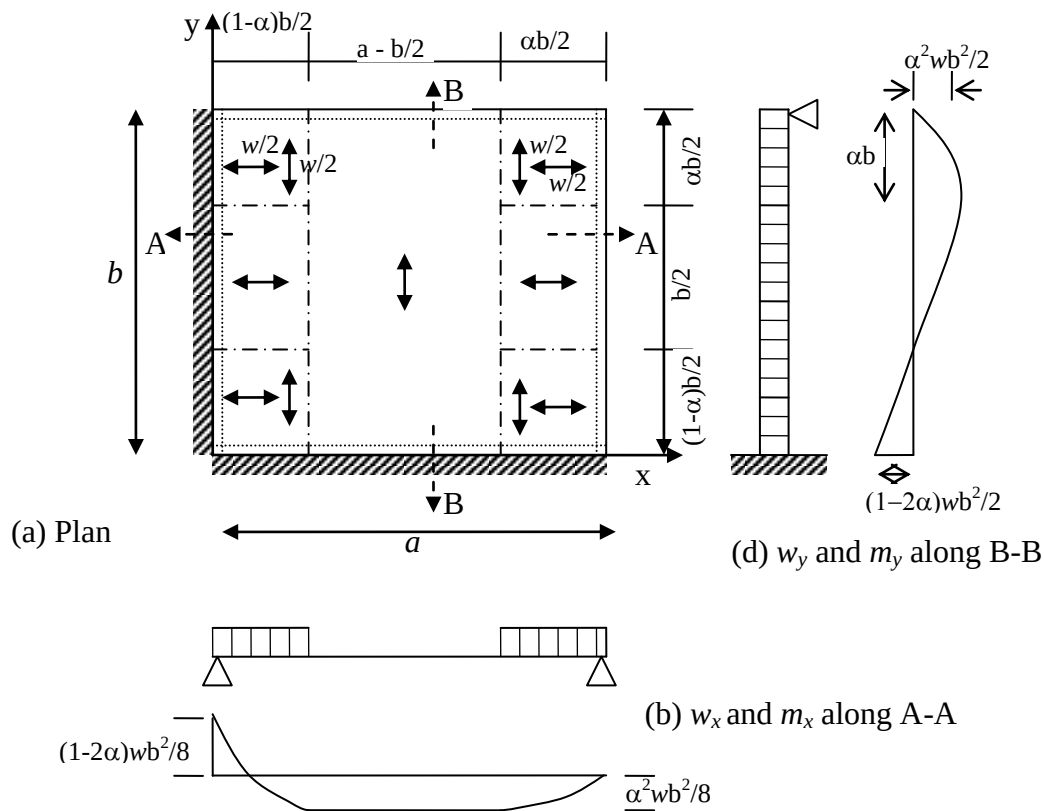
Fixed edges and continuity

Up to now we have dealt with positive moments in strips, where a large amount of flexibility in assigning loads to the various regions of the slab was provided. This same flexibility extends to the assignment of moments between negative and positive bending sections of slabs (strips) that are fixed or continuous over their supported edges. Some attention should be paid to elastic moment ratios to avoid problems with cracking and deflection at service loads.

The figure below shows a uniformly loaded rectangular slab having two adjacent fixed edges and the other two edges simply supported. Let us consider slab strips with one end fixed and one end simply supported as shown below. In determining by strip method, slab strips carrying loads only near the supports and unloaded in the central region are encountered (see figure). It is convenient if the unloaded region is subject to a constant moment (and zero shear) because this simplifies the selection of positive reinforcement.

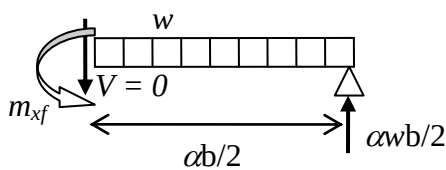
The following are recognized:

- Although the middle strips have the same width as those of the rectangular slab with simple supports, the discontinuity lines are shifted to account for the greater stiffness of the strips with fixed ends. Their location is defined by coefficient α , with a value clearly less than 0.5, so that the edge strips have widths greater and less than $b/4$ at the fixed end and simple end respectively (see fig.).

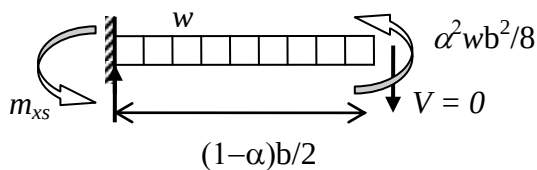


- For a BM diagram for x- direction middle strips (section A-A) with constant moment, over the unloaded part the following maximum moments are achieved.

Positive moment in the span



$$m_{xf} = w \left(\frac{ab}{2} \right)^2 - \frac{wab}{2} \left(\frac{1}{2} * \frac{ab}{2} \right) = \frac{\alpha^2 wb^2}{8}$$



Negative moment at the left support

$$m_{xs} = w(1-\alpha) \frac{b}{2} * (1-\alpha) \frac{b}{4} - \frac{\alpha^2 wb^2}{8}$$

$$= \frac{(1-2\alpha)wb^2}{8}$$

Observing, the absolute of the negative moment at a support plus the span moment = the "cantilever"

$$\text{moment.} = \frac{\alpha^2 wb^2}{8} + \frac{(1-2\alpha)wb^2}{8} = \frac{(1-\alpha)^2 wb^2}{8}$$

Now the ratio of negative to positive moments in the x-direction middle strip is:

$$\frac{m_{xs}}{m_{xf}} = \frac{1-2\alpha}{\alpha^2}$$

Hillerborg notes that as general rule for fixed edges, the support moment should be about 1.5 to 2.5 times the span moment in the same strip.

$$\Rightarrow \text{For } m_{xs}/m_{xf} = 2$$

$$2\alpha^2 + 2\alpha - 1 = 0 \Rightarrow \alpha = 0.366$$

Higher values should be chosen for longitudinal strips that are largely unloaded and in such cases a ratio of support to span moment of 3 to 4 may be used. However Asmin may govern for such high ratios with too small positive moment.

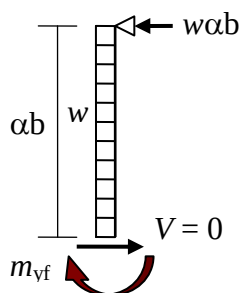
- Next moment in the x- direction edge strips:

Note that they are one half of those in the middle strips because load is half as great.

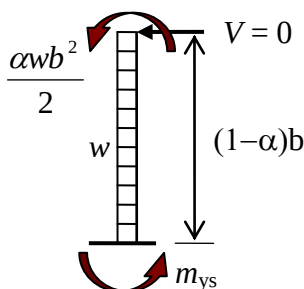
- Moment in the y- direction middle strips:

It is reasonable to choose the same ratio between support and span moments in the y- direction as in the x- direction.

→ Choose the distance from the right support to maximum moment section as αb [the cantilever span = $(1-\alpha)b \Rightarrow m_{ys} = (1-2\alpha)wb^2/2$]



$$m_{yf} = w\alpha b(\alpha b) - w\alpha b\left(\frac{\alpha b}{2}\right) = w\frac{(\alpha b)^2}{2} = \alpha^2 \frac{wb^2}{2}$$

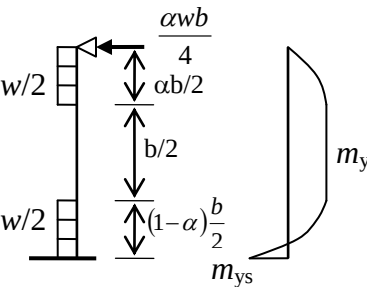


$$\begin{aligned} m_{ys} &= w(1-\alpha)b \cdot (1-\alpha)\frac{b}{2} - \alpha^2 \frac{wb^2}{2} \\ &= (1-\alpha)^2 \frac{wb^2}{2} - \alpha^2 \frac{wb^2}{2} \\ &= (1-2\alpha) \frac{wb^2}{2} \end{aligned}$$

Hence, the ratio of negative to positive moment is as before:

$$\frac{m_{xs}}{m_{xf}} = \frac{1-2\alpha}{\alpha^2}$$

Moment in the y-direction edge strips:



$$m_{yf} = \frac{w\alpha b}{2} \left(\frac{\alpha b}{2} \right) - \frac{w}{2} \frac{\alpha b}{2} \frac{\alpha b}{2} = \frac{w(\alpha b)^2}{16}$$

Cantilever moment

$$m_{ys} = \frac{w}{2} (1-\alpha) \frac{b}{2} \cdot (1-\alpha) \frac{b}{4} - \alpha^2 \frac{wb^2}{16}$$

$$= (1-\alpha)^2 \frac{wb^2}{16} - \alpha^2 \frac{wb^2}{16} = (1-2\alpha) \frac{wb^2}{16}$$

One-eighth of those in y- direction middle strip

With the above expressions, all the design moments for the slab can be found once a suitable value for α is chosen. $0.35 \leq \alpha \leq 0.39$ give corresponding ratios of negative to positive moments from 2.45 to 1.45, the range recommended by Hillerborg. For example, if it is decided that support moment is to be twice the span moments, the value of $\alpha = 0.366$ and the negative and positive moments in the central strip in the y- direction are respectively $0.134wb^2$ and $0.067wb^2$. In the middle strip in the x- directions, moments are one-fourth those values; and in the edge strips in both directions, they are one-eighth of those values.

Example

Figure below shows a typical interior panel of a slab floor in which support is provided by beams on all column lines. Hence the slab can be considered fully restrained on all sides. The floor must carry a live load of 6 kN/m^2 with C25 grade concrete and steel having $f_{yk} = 420 \text{ MPa}$. The dimensions of the slab panel are shown in the figure. Find the moment's at all critical sections and determine the required slab thickness and reinforcement.

Solution: Depth required for serviceability:

$$\text{Effective depth of slab} = d \geq \left(0.4 + \frac{0.6 f_{yk}}{400} \right) \frac{L_e}{\beta_a}$$

Here L_e = span of the joist = 6 m

β_a for slab span ratio 2:1 (for interior spans) = 35

β_a for slab span ratio 1:1 (for interior spans) = 45

β_a for slab span ratio 1.25:1 (Interpolated) = 42.5

$$d = (1.03) \frac{L_e}{\beta_a} = (1.03) \frac{6000}{42.5} = 145.4 \text{ mm}$$

Overall depth of the slab = $h = 145.4 + 15 + 12 = 172.4 \text{ mm}$,

Provide $h = 175 \text{ mm}$

Loads on the slab

DL of the slab = $(0.175 \times 25) = 4.375 \text{ kN/m}^2$

LL given in the problem = 6 kN/m^2

Design load = $1.3(4.375) + 1.6(6) = 15.29 \text{ kN/m}^2$

$W = 15.3 \text{ kN/m}^2$ $W/2 = 7.65 \text{ kN/m}^2$

Strips in the slabs

The discontinuity lines are selected as shown in the figure below.

Edge strip width = $b/4 = 6/4 = 1.5 \text{ m}$

In the corners the load is divided equally in the two directions; elsewhere 100 percent of load is assigned to the direction indicated by the arrows.

A ratio of support moment to the span moment of 2 is used

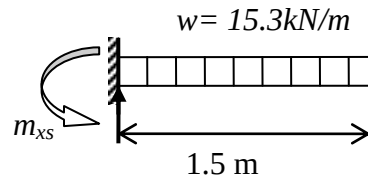
Calculation of moments

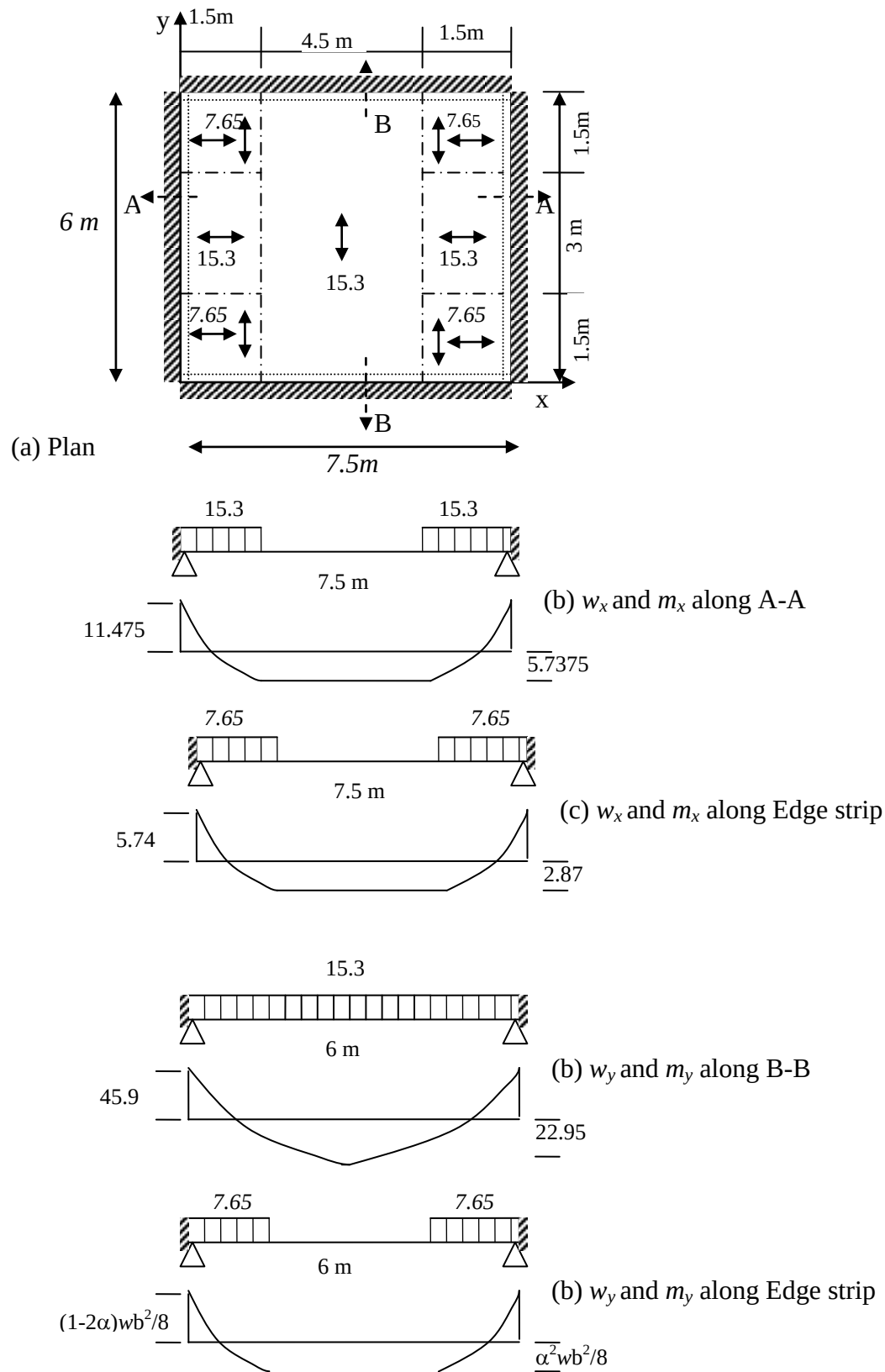
X direction middle strip along A-A:

Cantilever moment: $m_x = 15.3 * 1.5^2 / 2 = 17.21 \text{ kNm}$

Negative Moment: $m_{xs} = 17.21 * 2/3 = 11.475 \text{ kNm}$

Positive moment: $m_{xf} = 17.21 * 1/3 = 5.7375 \text{ k}$





X direction edge strip:

Cantilever moment: $m_x = 7.65 * 1.5^2 / 2 = 8.61 \text{ kNm}$

Negative Moment: $m_{xs} = 8.61 * 2/3 = 5.74 \text{ kNm}$

Positive moment: $m_{xf} = 8.61 * 1/3 = 2.87 \text{ kNm}$

Y direction middle strip along B-B:

Simply supported span moment: $m_y = 15.3 * 6^2 / 8 = 68.85 \text{ kNm}$

Negative Moment: $m_{ys} = 68.85 * 2/3 = 45.9 \text{ kNm}$

Positive moment: $m_{yf} = 68.85 * 1/3 = 22.95 \text{ kNm}$

Y direction edge strip:

Cantilever moment: $m_y = 7.65 * 1.5^2 / 2 = 8.61 \text{ kNm}$

Negative Moment: $m_{ys} = 8.61 * 2/3 = 5.74 \text{ kNm}$

Positive moment: $m_{yf} = 8.61 * 1/3 = 2.87 \text{ kNm}$

Desgn	Moment (kNm)	Depth chk (mm)	St ratio ρ	ρ_{min} condition	$A_s = \rho b d$ (mm ²)	Spacing (mm)	Sp. Prov (mm)	Dia of rod
X- midd								
$M_{xs} =$	11.475	53.46	0.001424	0.001424	213.54	235.41	230	8mm
$M_{xf} =$	5.7375	37.80	0.000705	0.00119	178.50	281.62	280	8mm
X- edge								
$M_{xs} =$	5.74	37.81	0.000705	0.00119	178.50	281.62	280	8mm
$M_{xf} =$	2.87	26.74	0.000351	0.00119	178.50	281.62	280	8mm
Y- midd								
$M_{ys} =$	45.9	106.92	0.006082	0.006082	912.36	123.96	120	12mm
$M_{yf} =$	22.95	75.61	0.002906	0.002906	435.94	259.44	250	12mm
Y- Edge								
$M_{ys} =$	5.74	37.81	0.000705	0.00119	178.50	281.62	280	8mm
$M_{yf} =$	2.87	26.74	0.000351	0.00119	178.50	281.62	280	8mm

Unsupported edges

Problems with unsupported edges could not be handled by conventional procedures so easily. The real power of the strip method becomes evident when dealing with non-standard problems, such as slabs with unsupported edge, slabs with holes, or slabs with reentrant edges (L - shaped slabs).

For a slab with one edge unsupported, a reasonable basis for analysis by the simple strip method is that a strip along the unsupported edge takes a greater load per unit area than the actual load acting, i.e., the strip along the unsupported edge acts as a support for the strips at right angles. Such strips have been referred to by Wood and Armer as "strong bands". A strong band is, in effect, an integral beam, usually having the same total depth as the remainder of the slab but containing a concentration of reinforcement. The strip may be made deeper than the rest of the slab to increase its carrying capacity, but this will not usually be necessary.

Slab with free edge in short span direction

Consider the rectangular slab carrying a uniformly distributed ultimate load per unit area, with fixed edges along three sides and no support along one short side.

Consider a strip along A-A in the x direction. Summing moments about the left end, with unknown support moment m_{xs} ,

$$m_{xs} + \frac{wb^2}{32} - \frac{kwb}{4} \left(a - \frac{b}{8} \right) = 0 \quad \text{From which, } k = \frac{1 + 32m_{xs} / wb^2}{8(a/b) - 1}$$

Thus k can be calculated after the support moment is selected.

The appropriate value of m_{xs} to be used in the above equation will depend on the shape of the slab. If a is large relative to b, the strong band in the y direction at the edge will be relatively stiff, and the moment in the left support in the x direction strips will approach the elastic value for a propped cantilever. If the slab is nearly square, the deflection of the strong band will tend to increase the support moment; a value about half the free cantilever moment shall be selected.

Once m_{xs} is selected and k value is known, it is easily shown that the maximum span moment occurs when

$$X = (1 - k) \frac{b}{4}$$

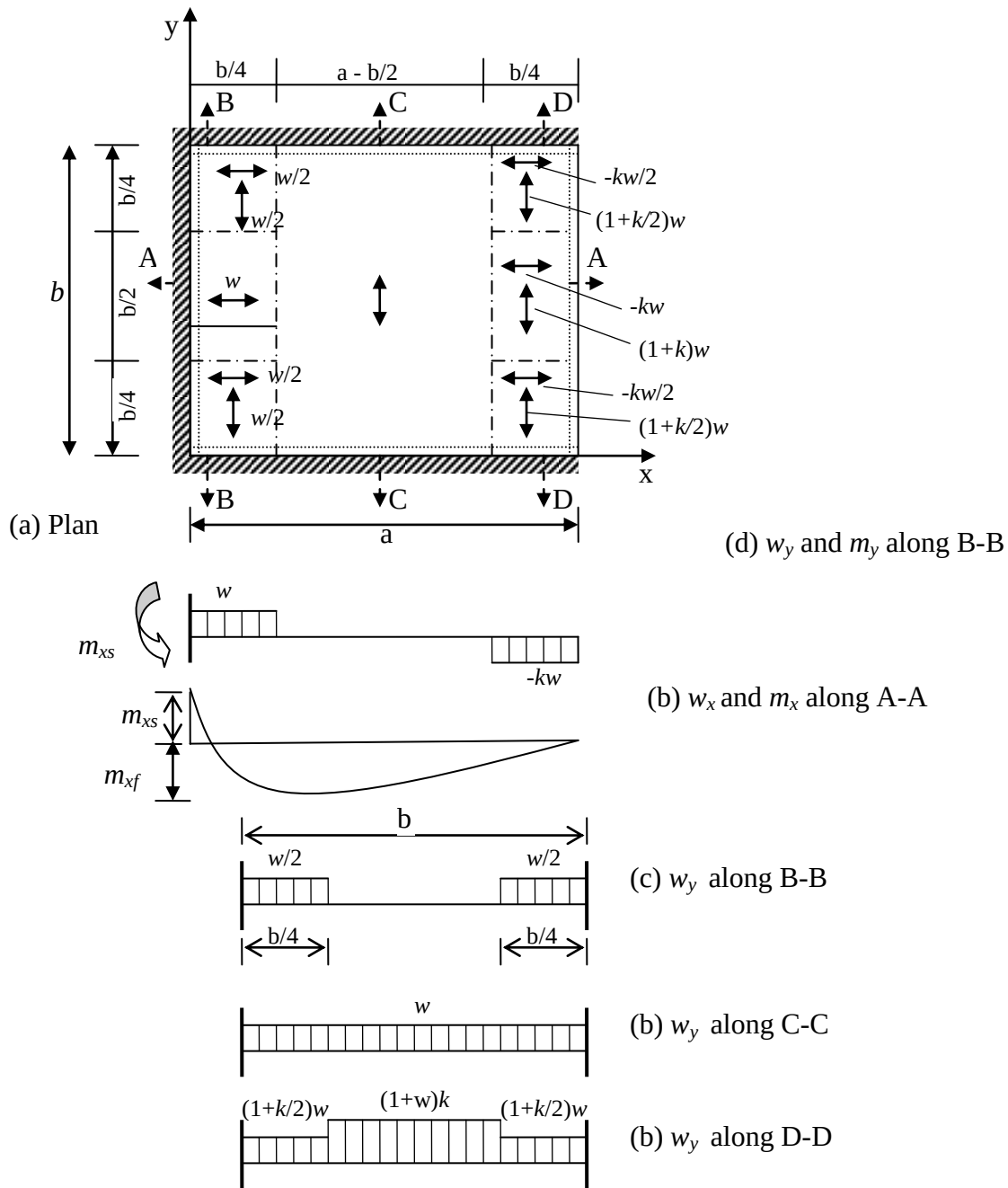


Fig. Slab with free edge along short side

And it has a value,

$$m_{xf} = \frac{kwb^2}{32} \left(\frac{8a}{b} - 3 + k \right)$$

The moments in the x direction edge strips are one-half of those in the middle strip.

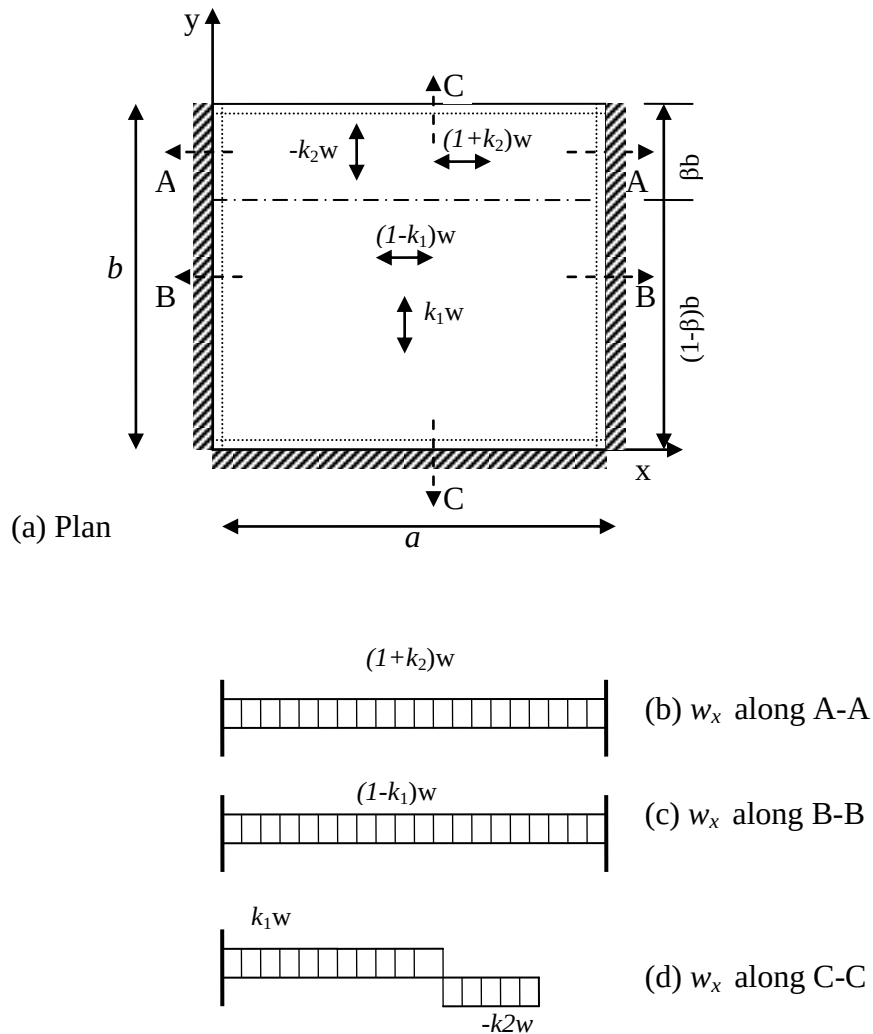
Y- direction middle strip along C-C: Simply supported span Moment = $wb^2/8$

Adopting a ratio of support to span moment of 2,

$$m_{xs} = \frac{wb^2}{12} \quad \& \quad m_{xf} = \frac{wb^2}{24}$$

moments along sections B-B and D-D can also be found by the same principles for the corresponding load values, with appropriate ratios of negative and positive moments

Slab with free edge in long span direction



Suitable discontinuity lines for the load distribution are shown in the figure below. Width of strong band along the free edge is βb (normally chosen as low as possible considering the limitations on tension reinforcement ratio in the strong band).

Moment equilibrium equation for Y direction strip =

$$m_{ys} + \frac{k_1 w (1 - \beta)^2 b^2}{2} - k_2 w \beta b^2 \left(1 - \frac{\beta}{2}\right) = 0$$

From which,

$$k_2 = \frac{k_1(1-\beta)^2 + 2m_{ys}/wb^2}{\beta(2-\beta)}$$

Having the value of k_1 selected, k_2 can be found.

Example 2

Rectangular slab with long edge unsupported

The 3.65 m x 5.8 m slab shown in the figure with three fixed edges and one long edge unsupported must carry a uniformly distributed service live load of 6 kN/m². Consider concrete grade to be used as C30 and steel to be used has $f_{yk} = 420$ MPa. Select an appropriate slab thickness, determine all factored moments in the slab, and select reinforcing bars and spacing for the slab.

Solution

Depth required for serviceability:

$$\text{Effective depth of slab} = d \geq \left(0.4 + \frac{0.6 f_{yk}}{400} \right) \frac{L_e}{\beta_a}$$

Here L_e = span of the slab = 3.65 m

β_a for slab span ratio 2:1 (for exterior spans) = 30

β_a for slab span ratio 1:1 (for exterior spans) = 40

Slab span ratio for the problem = $L_y/L_x = 5.8/3.65 = 1.59$

β_a for slab span ratio 1.59:1 (Interpolated) = 34.1

$$d = (1.03) \frac{L_e}{\beta_a} = (1.03) \frac{3650}{34.1} = 110.25 \text{ mm}$$

Overall depth of the slab = $h = 110.25 + 15 + 12 = 137.25$ mm,

Provide $h = 140$ mm

Loads on the slab

$$DL \text{ of the slab} = (0.14 * 25) = 3.5 \text{ kN/m}^2$$

$$LL \text{ given in the problem} = 6 \text{ kN/m}^2$$

$$\text{Design load} = 1.3(3.5) + 1.6(6) = 14.15 \text{ kN/m}^2$$

$$W = 14.15 \text{ kN/m}^2$$

Assumptions:

Width of strong band along the free edge = 0.65 m

In the main slab portion $k_1 = 0.45$

$$\text{Slab load in } y \text{ direction} = 0.45 * 14.15 = 6.37 \text{ kN/m}^2$$

$$\text{In } x \text{ direction} = 0.55 * 14.15 = 7.78 \text{ kN/m}^2$$

Y direction slab strip

As an initial assumption,

Negative moment at the supported edge = $\frac{1}{2}$ * the free cantilever moment

Consider the strip length as to span up to the center of the strong band;

$$\text{ie. } 3 + 0.65/2 = 3.325 \text{ m}$$

$$\text{Hence } m_{ys} = \frac{1}{2} * (6.37 * 3.325^2) / 2 = 17.6 \text{ kNm}$$

$$k_2 = \frac{k_1(1-\beta)^2 + 2m_{ys} / wb^2}{\beta(2-\beta)}$$

$$k_2 = \frac{0.45(1-0.65/3.65)^2 - 2 * 17.6 / 14.15 * 3.65^2}{0.65/3.65(2-0.65/3.65)} = 0.36$$

$$\text{Uplift in the strong band for } Y \text{ direction strips} = 0.36 * 14.15 = 5.095 \text{ kN/m}^2$$

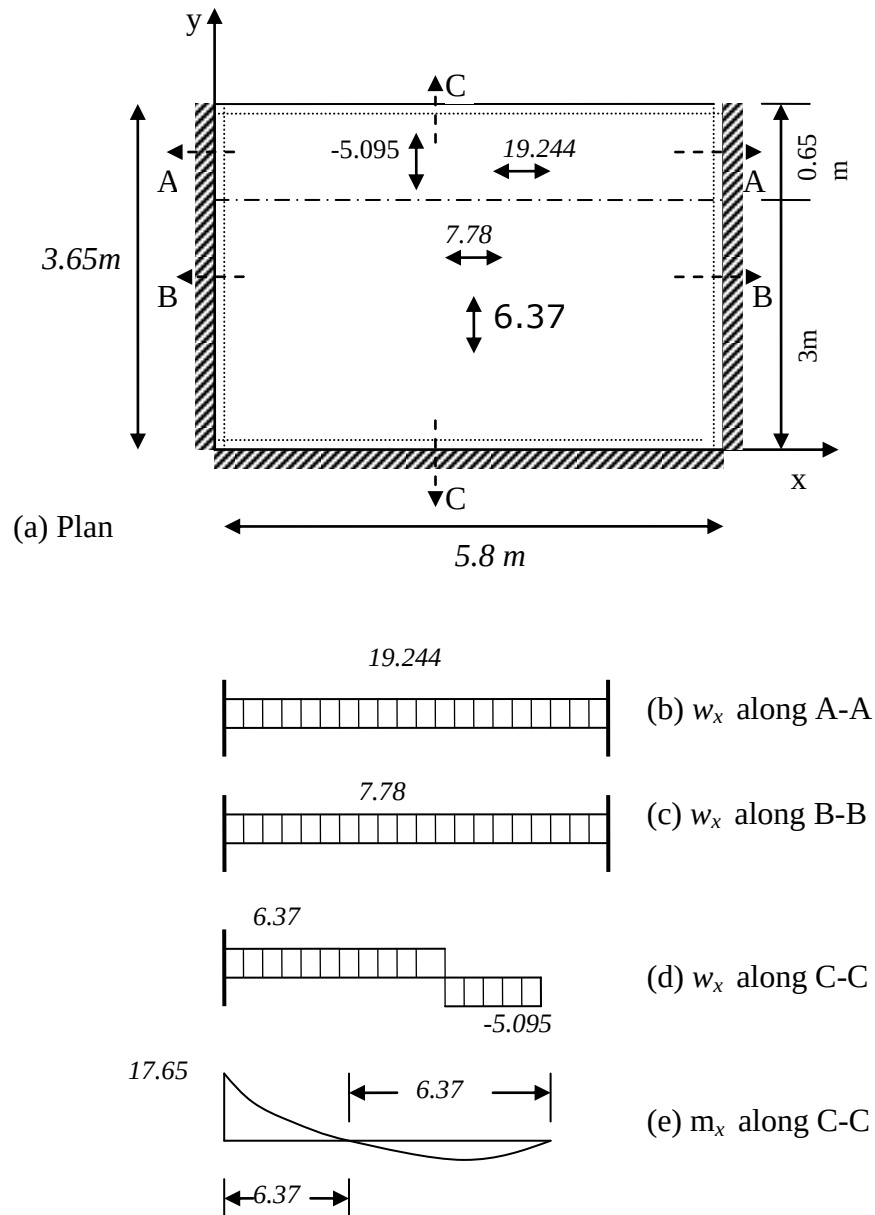
Revised negative moment at the left support

$$m_{ys} = 6.37 * 3^2 / 2 - 5.095 * 0.65 * 3.325 = 17.65 \text{ kNm}$$

The maximum positive moment in the Y direction strip will be located at the point of zero shear.

$$\text{ie. } 5.095 * 0.65 - 6.37(y_1 - 0.65) = 0$$

$$y_1 = 1.17 \text{ m}$$



Maximum Positive moment at ($y_1 = 1.17$ m)

$$= 5.095 * 0.65 * (1.17 - 0.65/2) - 6.37 * (1.17 - 0.65)^2/2 = 1.94 \text{ kNm}$$

X direction moments:

In strip along A-A:

$$\text{Load} = (1+k_2)w = (1+0.36) * 14.15 = 19.244 \text{ kN/m}^2$$

$$\text{Load per meter run along the strip} = 19.244 * 0.65 = 12.51 \text{ kN/m}$$

$$\text{Simply supported span moment} = 12.51 * 5.8^2/8 = 52.6 \text{ kNm}$$

$$\text{Negative moment at the supports} = (2/3) * 52.6 = 35.1 \text{ kNm}$$

$$\text{Positive moment at the span} = (1/3) * 52.6 = 17.5 \text{ kNm}$$

In strip along B-B:

$$\text{Load per meter run along the strip} = 7.78 \text{ kN/m}$$

$$\text{Simply supported span moment} = 7.78 * 5.8^2/8 = 32.71 \text{ kNm}$$

$$\text{Negative moment at the supports} = (2/3) * 32.71 = 21.8 \text{ kNm}$$

$$\text{Positive moment at the span} = (1/3) * 32.71 = 10.9 \text{ kNm}$$

Slabs with holes

Example:

A rectangular slab, 5 m x 8 m with fixed supports at all the four sides has a central opening of 1.2 m x 2.4 m. Slab thickness is Calculated to be of 200mm. The slab is to carry a uniformly distributed factored load of 15 kN/m² including its self weight. Device an appropriate system of strong bands to reinforce the opening and determine moments to be resisted at all critical sections of the slab.

Solution

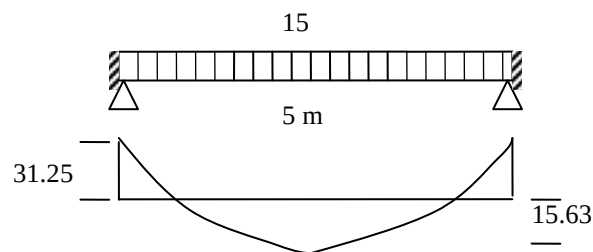
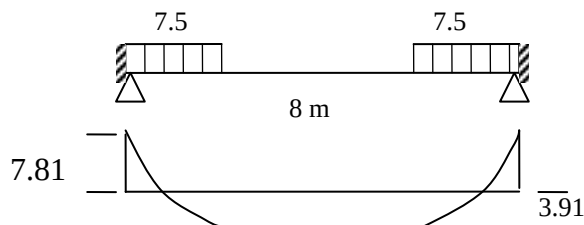
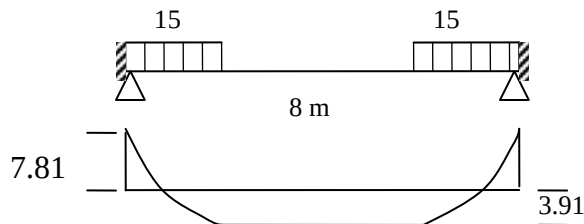
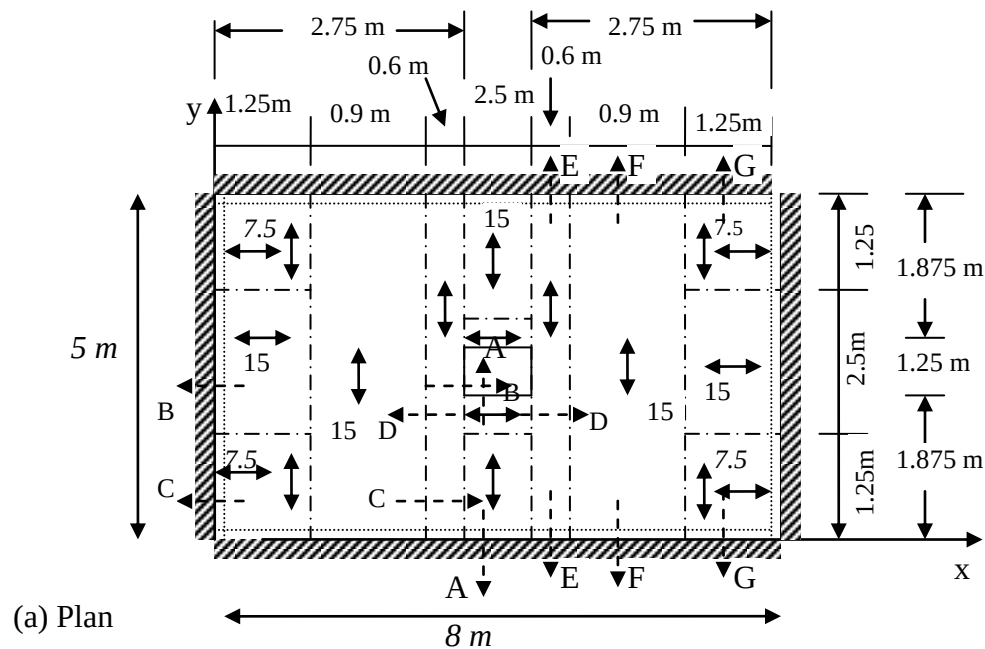
Moments for slab without holes

$$\text{X direction middle strips: } w = 15 \text{ kN/m}^2$$

$$\text{Cantilever moment: } m_x = 15 * 1.25^2/2 = 11.72 \text{ kNm}$$

$$\text{Negative Moment: } m_{xs} = 11.72 * 2/3 = 7.81 \text{ kNm}$$

$$\text{Positive moment: } m_{xf} = 11.72 * 1/3 = 3.91 \text{ kNm}$$



X direction edge strips: $w = 7.5 \text{ kN/m}^2$

Cantilever moment: $m_x = 7.5 * 1.25^2 / 2 = 5.86 \text{ kNm}$

Negative Moment: $m_{xs} = 5.86 * 2/3 = 3.91 \text{ kNm}$

Positive moment: $m_{xf} = 5.86 * 1/3 = 1.95 \text{ kNm}$

Y direction middle strips: $w = 15 \text{ kN/m}^2$

Simply supported span moment: $m_x = 15 * 5^2 / 8 = 46.88 \text{ kNm}$

Negative Moment: $m_{xs} = 46.88 * 2/3 = 31.25 \text{ kNm}$

Positive moment: $m_{xf} = 46.88 * 1/3 = 15.63 \text{ kNm}$

Y direction edge strips: $w = 7.5 \text{ kN/m}^2$

Cantilever moment: $m_x = 7.5 * 1.25^2 / 2 = 5.86 \text{ kNm}$

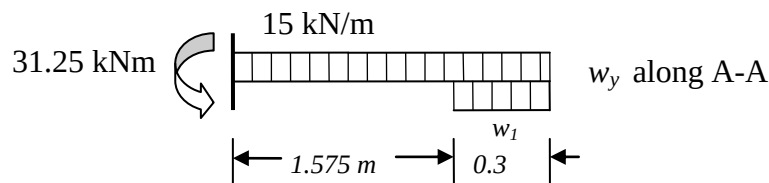
Negative Moment: $m_{xs} = 5.86 * 2/3 = 3.91 \text{ kNm}$

Positive moment: $m_{xf} = 5.86 * 1/3 = 1.95 \text{ kNm}$

Because of the hole, certain strips lack support at one end. To support them 0.3m wide strong bands will be provided in the X direction at the long edges of the hole and 0.6m wide strong bands in the Y direction at the short edges of the hole.

Strip A-A

Assuming propped cantilever action with restraint moment along the slab edge, taken as $m_{ys} = 31.25 \text{ kNm}$ as in the basic case,



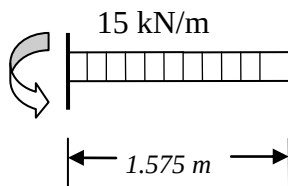
By moment equilibrium about the left support,

$$31.25 + w_1 * 0.3 * (1.575 + 0.3/2) - 15 * (1.875)^2 / 2 = 0$$

$$w_1 = - 9.44 \text{ kN/m}$$

The negative value of w_1 indicates that the cantilever strips are serving as support for strip D-D and in turn for the strong bands in the Y direction, which is hardly a reasonable assumption. Hillerborg suggests the restraint moment to be as close to the "basic case" as possible without w_1 being negative i.e. choosing $w_1 = 0$ (cantilever alone).

$$m_{ys} = 15 * (1.575)^2 / 2 = 18.6 \text{ kNm}$$



w_y along A-A

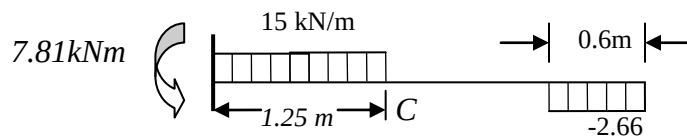
18.6 kNm



m_{ys} at the support in
Strip A-A

Strip B-B

The restraint moment at the support from the basic case = 7.81kNm



(b) w_x and m_x along B-B



Summing moments about the left end of the strip, results in an uplift reaction at the right end to be provided by strip E-E.

$$7.81 + w_2 * 0.6 * (1.25 + 0.9 + 0.6/2) - 15 * (1.25)^2 / 2 = 0$$

$$W_2 = - 2.66 \text{ kN/m}$$

Taking moment about C,

$$\begin{aligned}\text{Left support reaction} &= \{7.81 + 15 * 1.25^2/2\} / 1.25 \\ &= 15.623 \text{ kN}\end{aligned}$$

Point of zero shear: $15.623 - 15x = 0$

$$X = 15.623/15 = 1.042 \text{ m}$$

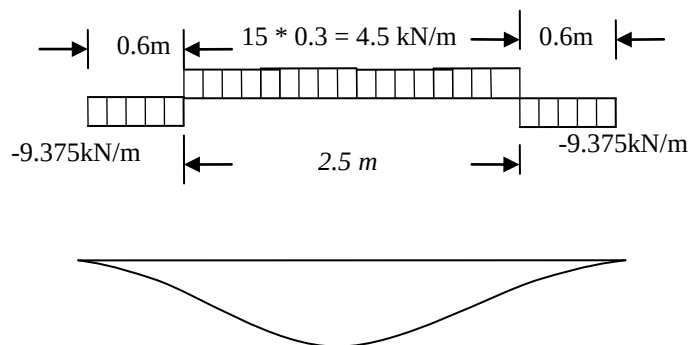
$$\begin{aligned}\text{Max. BM in the span} &= 15.623 * 1.042 - 7.81 - 15 * 1.042^2/2 \\ &= 0.3259 \text{ kNm}\end{aligned}$$

Strip C-C

BM values for strip C-C are half of the corresponding values for strip B-B

Strip D-D

The 0.3m width strip D-D carries 15 kN/m in the X direction with reactions provided by the strong bands E-E.



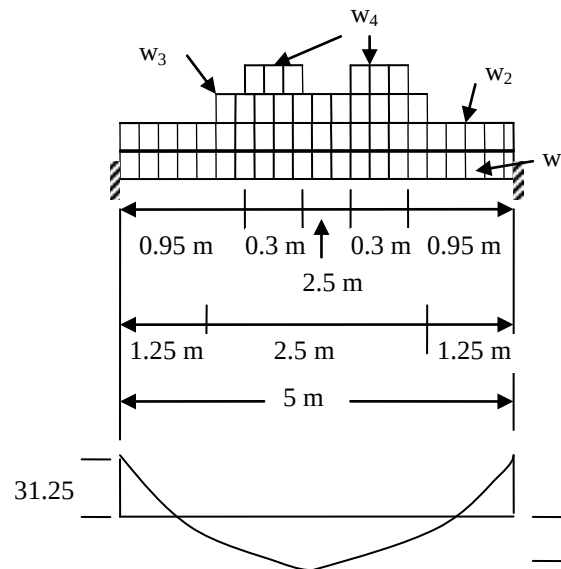
$$\begin{aligned}\text{Reaction on E-E} &= \frac{1}{2} * 4.5 * 2.5/0.6 = 9.375 \text{ kN/m} \\ &\text{or } 9.375/0.3 = 31.25 \text{ kN/m}^2\end{aligned}$$

The maximum +ve moment is

$$M_{xf} = 0.6 * 9.375 * 1.55 - 4.5 * 1.25^2 / 2 = 5.2 \text{ kNm}$$

Strip E-E

The strong bands in the Y direction (the strips along section E-E) carry the directly applied load of 15 kN/m^2 plus the 2.66 kN/m^2 load from strip B-B, the 1.33 kN/m^2 load from the strip C-C, and the 31.25 kN/m^2 end reaction from strip D-D.



All the above mentioned loads are converted to kN/m loads by multiplying the width of the strip as follows and are indicated in the figure above also.

$$w_1 = 15 * 0.6 = 9 \text{ kN/m}$$

$$w_2 = 1.33 * 0.6 = 0.798 \text{ kN/m}$$

$$w_3 = (2.66 - 1.33) * 0.6 = 0.798 \text{ kN/m}$$

$$w_4 = 31.25 * 0.6 = 18.75 \text{ kN/m}$$

Cantilever moment:

$$9 * 2.5 * 1.25 + 0.798 * 2.5 * 1.25 + 0.798 * 1.25 * (1.25 + 0.625) + 18.75 * 0.3 * (0.95 + 0.15) = 38.68 \text{ kNm}$$

Negative moment: $38.68 * 2/3 = 25.78 \text{ kNm}$

Positive moment: $38.68 * 1/3 = 12.89 \text{ kNm}$

Strip F-F

The moments for the Y direction middle strip of the basic case (without hole) may be used with out change.

Strip G-G

The moments for the Y direction edge strip of the basic case (without hole) may be used with out change.